Srinivas Reddy Guntaka

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.70
Sri Chaitanya Junior College	Class XII (Telangana State Board of Intermediate Education)	2024	97.30%
Sri Chaitanya School	Class X (Board of Secondary Education, Telangana State)	2022	9.70%

SKILLS

Programming Languages: C, C++, Java, Python, JavaScript

Web & Database: HTML5, CSS3, Node.js, PostgreSQL, MySQL

Machine Learning & Data Science: PyTorch, NumPy, Pandas, Matplotlib

Graphics & UI: OpenGL, Dear ImGui

Developer Tools & Systems: Git, Docker, Unix/Linux, CMake, Make, GDB

Assembly: RISC-V

RELEVANT COURSES

Computer Science: Introduction to Programming, Data Structures, Algorithms, Computer Architecture, Operating Systems I & II, Compilers-1, Database Management Systems I & II, Software Development Fundamentals, Theory of Computation, Introduction to AI

Mathematics: Discrete Mathematics for Computer Science, Probability and Random Variables, Linear Algebra, Introduction to Statistics, Transform Techniques

PROJECTS

Configurable RISC-V CPU Simulator | GitHub | *C++, OpenGL, Dear ImGui, CMake*

Developed a **26k+ LOC** modular architectural simulation framework to model instruction-level parallelism, resource constraints, and hardware hazard resolution.

- **Simulation Engine:**
- **Architected execution backends** spanning single-cycle, 5-stage in-order pipelined, and ***dynamic superscalar variants (dual and triple-issue)***.
- **Engineered a deterministic state-rollback system** allowing full, cycle-by-cycle "undo" capabilities by capturing and restoring microarchitectural snapshots (registers, PC, and memory state).
- **Implemented microarchitectural abstractions** for dynamic scheduling, including reservation stations, a dedicated register status file, and a commit-buffer to preserve program order during multi-instruction completion.
- **Graphics & Analytics UI:**
- **Developed an interactive frontend** using OpenGL and Dear ImGui to visually profile and debug pipeline occupancy, execution latency, CPI, and data/structural hazards in real time.
- **Built upon a custom assembly front-end** featuring text tokenization, bitwise opcode masking (I/S/B/U/J formats), and lifecycle metadata tracking to stream simulation states seamlessly to the UI.

Neural Associative Content-Addressable Memory System | GitHub | *Python, PyTorch, CNNs, Attention, NumPy*

Developed an external, non-parametric Content-Addressable Memory (CAM) framework to store images and retrieve

them deterministically using heavily degraded or partial visual cues.

- **Pipeline & Architecture:**
- **Engineered a decoupled, 3-stage pipeline** featuring a Convolutional Autoencoder for feature reduction and a custom CueEncoder trained to isolate structural signals from 25%-60% pixel erasure and block masking.
- **Mapped high-dimensional spatial features** into optimized 512-dimensional latent context vectors across CIFAR and STL-10 datasets.
- **Memory & Retrieval:**
- **Implemented a neural Comparator** using MultiheadAttention to execute soft and hard (winner-takes-all) lookups against stored memory banks.
- **Optimized retrieval convergence** using a joint loss function, combining dual-head Cross-Entropy for memory selection and MSE for pixel-level reconstruction.

ADDITIONAL PROJECTS

Custom C++ Templates Library: Developed a native utility library from scratch implementing common data structures (vectors, hash maps, priority queues etc) leveraging advanced C++ templates for generic type safety.

Infinite-Precision Java Library: Built an arbitrary-precision arithmetic library capable of handling scaling numerical values, implementing the Karatsuba algorithm for optimized large-integer multiplication and and Newton-Raphson division.

Voronoi Terrain Generator: Programmed a procedural terrain generation application using Voronoi diagrams and computational geometry algorithms to dynamically map out structural cell regions.

CODING PROFILES

LeetCode: Levistus71 -- **845+ Problems Solved, Ranked** 53,834